

Probability



Basic probability

- 1 Two fair six-sided dice are rolled. Find the probability their sum is 8.
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Complement rule

- 2 If $P(E) = 0.35$, find $P(E')$.
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Permutations and combinations

- 3 How many ways can a committee of 4 be chosen from 10 people?
- 4 How many ways can 2 officers (president, vice president — order matters) be chosen from 6 candidates?
- 5 Explain, using Theorem 6.1, how $\binom{6}{2}$ relates to $P(6, 2)$ from the previous question.
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Independent events and the addition rule

- 6 Events A and B are independent, with $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.
- 7 A card is drawn from a standard 52-card deck. Find $P(\text{Queen or Spade})$.
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Multiple choice — easy

- 8 A bag contains 9 yellow marbles, 5 red marbles, and 7 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.
- a. $\frac{1}{3}$
- b. $\frac{3}{7}$
- c. $\frac{10}{21}$
- d. $\frac{5}{21}$

- 9 A bag contains 3 green marbles, 7 yellow marbles, and 7 red marbles. A marble is drawn at random. Find $P(\text{green})$.
- a. $\frac{7}{17}$
 - b. $\frac{5}{17}$
 - c. $\frac{4}{17}$
 - d. $\frac{3}{17}$
- 10 A bag contains 9 yellow marbles, 9 red marbles, and 6 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.
- a. $\frac{5}{12}$
 - b. $\frac{11}{24}$
 - c. $\frac{1}{4}$
 - d. $\frac{3}{8}$
- 11 A bag contains 5 yellow marbles, 3 red marbles, and 7 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.
- a. $\frac{7}{15}$
 - b. $\frac{2}{5}$
 - c. $\frac{1}{3}$
 - d. $\frac{1}{5}$
- 12 A bag contains 7 blue marbles, 7 green marbles, and 3 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.
- a. $\frac{9}{17}$
 - b. $\frac{8}{17}$
 - c. $\frac{3}{17}$
 - d. $\frac{7}{17}$

- 13 A bag contains 5 yellow marbles, 4 red marbles, and 4 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.
- a. $\frac{5}{13}$
 - b. $\frac{4}{13}$
 - c. $\frac{6}{13}$
 - d. $\frac{7}{13}$
- 14 A bag contains 7 blue marbles, 7 green marbles, and 5 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.
- a. $\frac{7}{19}$
 - b. $\frac{8}{19}$
 - c. $\frac{9}{19}$
 - d. $\frac{5}{19}$
- 15 A bag contains 2 blue marbles, 4 green marbles, and 4 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.
- a. $\frac{2}{5}$
 - b. $\frac{1}{5}$
 - c. $\frac{3}{10}$
 - d. $\frac{1}{2}$
- 16 A bag contains 4 blue marbles, 5 green marbles, and 5 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.
- a. $\frac{2}{7}$
 - b. $\frac{1}{2}$
 - c. $\frac{3}{7}$
 - d. $\frac{5}{14}$

- 17 A bag contains 3 yellow marbles, 3 red marbles, and 2 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.
- a. $\frac{1}{2}$
 - b. $\frac{1}{4}$
 - c. $\frac{5}{8}$
 - d. $\frac{3}{8}$
- 18 A bag contains 9 blue marbles, 2 green marbles, and 4 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.
- a. $\frac{4}{15}$
 - b. $\frac{2}{15}$
 - c. $\frac{2}{3}$
 - d. $\frac{3}{5}$
- 19 A bag contains 5 green marbles, 3 yellow marbles, and 3 red marbles. A marble is drawn at random. Find $P(\text{green})$.
- a. $\frac{6}{11}$
 - b. $\frac{3}{11}$
 - c. $\frac{7}{11}$
 - d. $\frac{5}{11}$
- 20 A bag contains 3 red marbles, 3 blue marbles, and 4 green marbles. A marble is drawn at random. Find $P(\text{red})$.
- a. $\frac{3}{10}$
 - b. $\frac{1}{2}$
 - c. $\frac{3}{5}$
 - d. $\frac{2}{5}$

- 21 A bag contains 6 red marbles, 8 blue marbles, and 9 green marbles. A marble is drawn at random. Find $P(\text{red})$.
- a. $\frac{6}{23}$
 - b. $\frac{7}{23}$
 - c. $\frac{9}{23}$
 - d. $\frac{8}{23}$
- 22 A bag contains 5 green marbles, 9 yellow marbles, and 3 red marbles. A marble is drawn at random. Find $P(\text{green})$.
- a. $\frac{3}{17}$
 - b. $\frac{9}{17}$
 - c. $\frac{5}{17}$
 - d. $\frac{6}{17}$
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Multiple choice — medium

- 23 How many ways can a committee of 6 be chosen from 7 people?
- a. 1
 - b. 3
 - c. 11
 - d. 7
- 24 How many ways can a committee of 5 be chosen from 6 people?
- a. 10
 - b. 1
 - c. 2
 - d. 6

- 25 How many ways can a committee of 5 be chosen from 16 people?
- a. 4368
 - b. 8008
 - c. 4364
 - d. 4372
- 26 How many ways can a committee of 4 be chosen from 11 people?
- a. 326
 - b. 334
 - c. 462
 - d. 330
- 27 How many ways can a committee of 3 be chosen from 14 people?
- a. 364
 - b. 368
 - c. 1001
 - d. 360
- 28 How many ways can a committee of 2 be chosen from 20 people?
- a. 194
 - b. 190
 - c. 186
 - d. 1140

- 29 How many ways can a committee of 3 be chosen from 10 people?
- a. 120
 - b. 116
 - c. 124
 - d. 210
- 30 How many ways can a committee of 2 be chosen from 19 people?
- a. 175
 - b. 167
 - c. 969
 - d. 171
- 31 How many ways can a committee of 4 be chosen from 9 people?
- a. 126
 - b. 122
 - c. 130
 - d. 132
- 32 How many ways can a committee of 3 be chosen from 15 people?
- a. 1365
 - b. 459
 - c. 455
 - d. 451

- 33 How many ways can a committee of 5 be chosen from 17 people?
- a. 6192
 - b. 12376
 - c. 6188
 - d. 6184
- 34 How many ways can a committee of 2 be chosen from 18 people?
- a. 153
 - b. 149
 - c. 157
 - d. 816
- 35 How many ways can a committee of 6 be chosen from 18 people?
- a. 18564
 - b. 31824
 - c. 18560
 - d. 18568
- 36 How many ways can a committee of 2 be chosen from 14 people?
- a. 87
 - b. 91
 - c. 364
 - d. 95

37 How many ways can a committee of 5 be chosen from 11 people?

- a. 458
- b. 462
- c. 468
- d. 466

Multiple choice — hard

38 How many ways can 2 distinct positions be filled from 12 candidates, where order matters?

- a. 66
- b. 138
- c. 132
- d. 126

39 How many ways can 4 distinct positions be filled from 9 candidates, where order matters?

- a. 3024
- b. 126
- c. 3018
- d. 3030

40 How many ways can 4 distinct positions be filled from 14 candidates, where order matters?

- a. 24018
- b. 1001
- c. 24030
- d. 24024

- 41 How many ways can 2 distinct positions be filled from 11 candidates, where order matters?
- a. 110
 - b. 55
 - c. 116
 - d. 104
- 42 How many ways can 4 distinct positions be filled from 15 candidates, where order matters?
- a. 32760
 - b. 32766
 - c. 32754
 - d. 1365
- 43 Events A and B are independent, with $P(A) = 0.1$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.
- a. $\frac{1}{2}$
 - b. $\frac{9}{100}$
 - c. $\frac{19}{100}$
 - d. $\frac{1}{25}$
- 44 Events A and B are independent, with $P(A) = 0.2$ and $P(B) = 0.3$. Find $P(A \text{ and } B)$.
- a. $\frac{1}{2}$
 - b. $\frac{1}{100}$
 - c. $\frac{11}{100}$
 - d. $\frac{3}{50}$

45 Events A and B are independent, with $P(A) = 0.5$ and $P(B) = 0.1$. Find $P(A \text{ and } B)$.

a. $\frac{1}{5}$

b. $\frac{3}{5}$

c. $\frac{1}{10}$

d. $\frac{1}{20}$

46 Events A and B are independent, with $P(A) = 0.2$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

a. $\frac{13}{100}$

b. $\frac{3}{100}$

c. $\frac{2}{25}$

d. $\frac{3}{5}$

47 Events A and B are independent, with $P(A) = 0.6$ and $P(B) = 0.3$. Find $P(A \text{ and } B)$.

a. $\frac{23}{100}$

b. $\frac{13}{100}$

c. $\frac{9}{50}$

d. $\frac{9}{10}$

48 Out of 38 equally likely tiles, 5 are red, 8 are numbered, and 3 are both. One item is chosen at random. Find $P(\text{red or numbered})$.

a. $\frac{5}{19}$

b. $\frac{6}{19}$

c. $\frac{13}{38}$

d. $\frac{4}{19}$

- 49 Out of 27 equally likely boxes, 7 are labeled, 8 are sealed, and 7 are both. One item is chosen at random. Find $P(\text{labeled or sealed})$.
- a. $\frac{2}{9}$
 - b. $\frac{10}{27}$
 - c. $\frac{8}{27}$
 - d. $\frac{5}{9}$
- 50 Out of 24 equally likely boxes, 4 are labeled, 8 are sealed, and 3 are both. One item is chosen at random. Find $P(\text{labeled or sealed})$.
- a. $\frac{11}{24}$
 - b. $\frac{3}{8}$
 - c. $\frac{7}{24}$
 - d. $\frac{1}{2}$
- 51 Out of 26 equally likely tickets, 9 are gold, 12 are striped, and 8 are both. One item is chosen at random. Find $P(\text{gold or striped})$.
- a. $\frac{21}{26}$
 - b. $\frac{15}{26}$
 - c. $\frac{1}{2}$
 - d. $\frac{11}{26}$
- 52 Out of 37 equally likely numbers, 4 are even, 10 are prime, and 2 are both. One item is chosen at random. Find $P(\text{even or prime})$.
- a. $\frac{14}{37}$
 - b. $\frac{10}{37}$
 - c. $\frac{12}{37}$
 - d. $\frac{15}{37}$